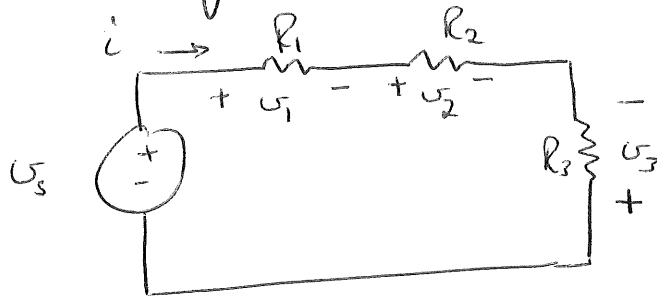


lesing 2.3

①

⇒ Series Resistors en Spanningsverdeling.

Om gebruik KVL om te zien hoe stroom en spanning in een serie kringbaar optree.



* Ohm se Wet

$$U_1 = +iR_1$$

$$U_2 = +iR_2$$

$$U_3 = -iR_3.$$

* KVL

$$-U_s + U_1 + U_2 - U_3 = 0$$

$$U_s = U_1 + U_2 - U_3$$

$$U_s = (iR_1) + (iR_2) - (-iR_3)$$

$$U_s = i(R_1 + R_2 + R_3)$$

Totale
stroom in
serie kringbaar

$$\Rightarrow i = \frac{U_s}{R_1 + R_2 + R_3}.$$

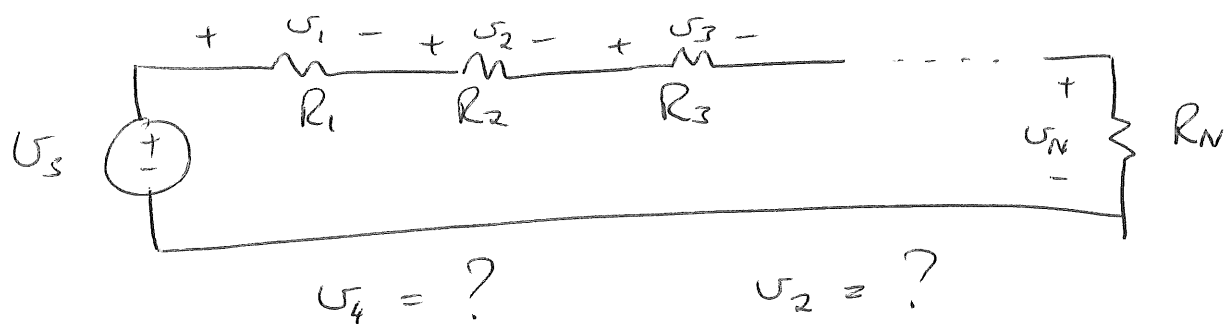
$$\rightarrow U_1 = iR_1 = \frac{U_s \cdot R_1}{R_1 + R_2 + R_3} = \frac{U_s R_1}{R_{eq.}}$$

$$\rightarrow U_3 = -iR_3 = \frac{-U_s R_3}{R_1 + R_2 + R_3} = \frac{-U_s R_3}{R_{eq}}$$

(2)

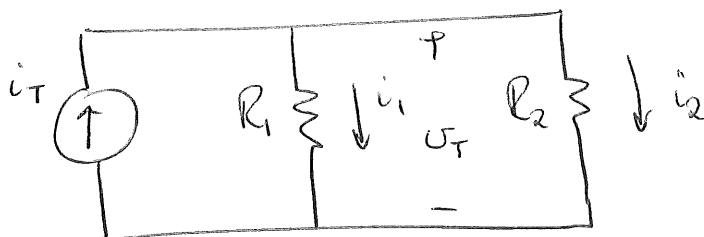
→ Series Resistors: $R_{eq} = \sum_{i=1}^N R_i$

→ Spanningverdeling $V_n = \frac{V_s}{R_{eq}} \cdot R_n$



⇒ Parallel Resistors en Stroomverdeling.

Ons gebruikt KCL om te zien hoe stroom en spanning in 'n parallel kringbaan optree.



* Ohm se Wet

$$i_1 = \frac{+V_T}{R_1}$$

$$i_2 = \frac{+V_T}{R_2}$$

* KCL

$$i_T = \frac{V_T}{R_1} + \frac{V_T}{R_2} = V_T \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\text{maar } i_T = \frac{V_T}{R_{eq}}$$

$$\therefore \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{eq}} = \frac{R_1 + R_2}{R_1 R_2}$$

2 Resistors

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = R_{||}$$

Totale spanning

$$\rightarrow U = i_T \times R_{eq}$$

* Alle elemente is parallel:

$$U_T = U_1 = U_2$$

$$\therefore U_T = i_T R_{eq} = i_1 R_1$$

$$\frac{i_T R_1 R_2}{R_1 + R_2} = i_1 R_1$$

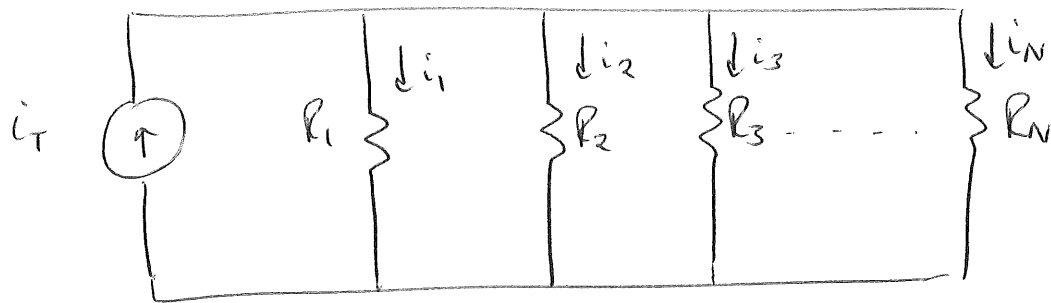
$$i_1 = \frac{i_T R_2}{R_1 + R_2}$$

$$i_2 = \frac{i_T R_1}{R_1 + R_2}$$

Stroomverdeling vir 2 resistors.

Wat gebeur by 3 en meer resistors?

(4)



$$i_T = \frac{U_T}{R_{eq}} = U_T \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_N} \right)$$

$$\therefore U_T = i_T \cdot R_{eq} = i_T \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_N} \right)^{-1}$$

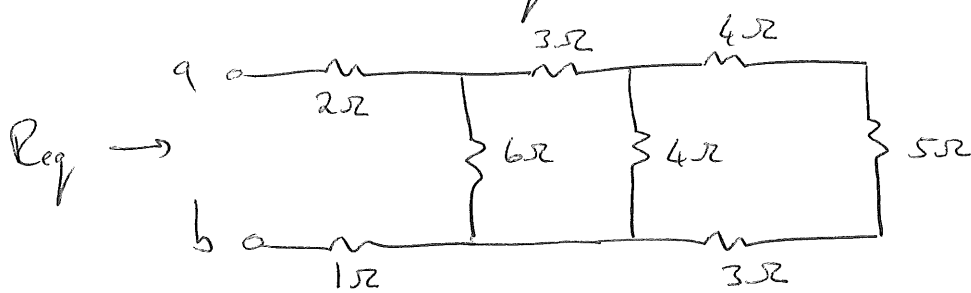
$$\rightarrow i_1 = \frac{U_T}{R_1}$$

$$i_2 = \frac{U_T}{R_2}$$

$$i_5 = ?$$

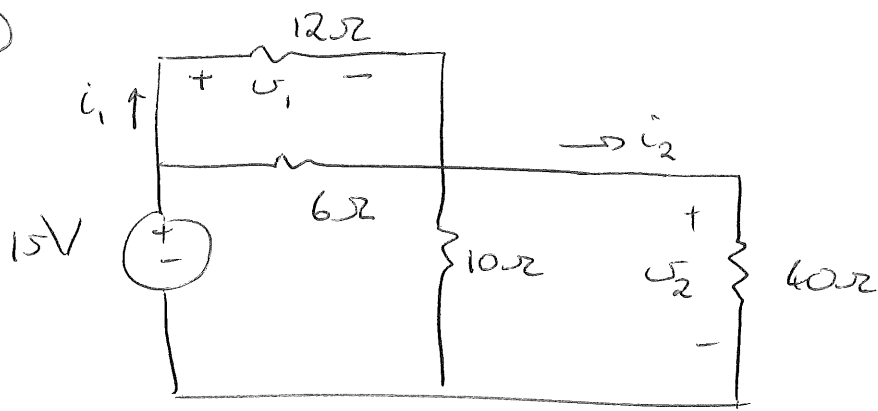
V6

①

Bereken R_{eq} .

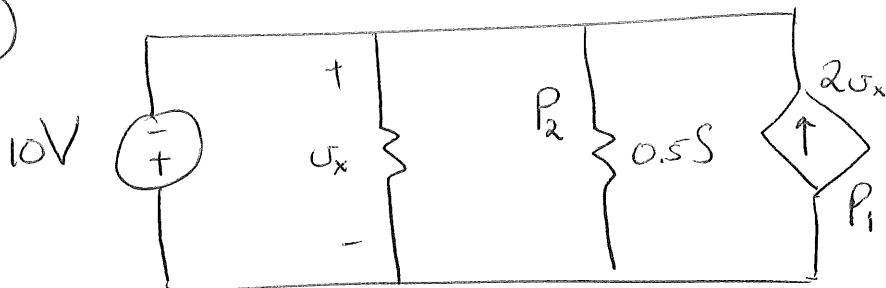
⑤

②



$$\begin{aligned} U_1 &= ? \\ U_2 &= ? \\ i_1 &= ? \\ i_2 &= ? \end{aligned}$$

③



$$\begin{aligned} P_2 &= ? \\ P_1 &= ? \end{aligned}$$

④

a) $V_s = 30V$

$i_y = ?$ $v_y = ?$

b) $i_T = 10mA$

$v_x = ?$

⑥

